GhostScan: scalable, self-validating detection of driven variables in large dynamical systems

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 Code, experiment ledger, pre-registration documents and figure sources: https://github.com/AkandaAshraf/DeepFeatSelection

Abstract

We address the problem of determining which variables of a large dynamical system are driven by the system and which evolve autonomously. Existing causal-detection methods cannot address this question at scale. We measured full pairwise convergent cross mapping (CCM) at $\sim$1.1 s per pair under the full protocol ($\approx$31,700 compute-hours for one scan of 10^4 variables); k-nearest-neighbour (kNN) conditioning became sample-starved at 18 conditioning dimensions in our two-seed synthetic tests; and PCMCI (the Peter–Clark algorithm with momentary conditional independence) with partial correlation produced mediated false positives near synchrony on synthetic chains. We first isolate three mechanisms by which *learned* readouts fail on such systems: maturity collapse of difference-based importance under Takens redundancy, typicality substitution in masked-reconstruction scores, and the inability of ensembling to remove shared bias. Each is demonstrated with a separating experimental signature. These mechanisms jointly imply a corrected statistic, an amortised Granger/transfer-entropy-type quantity we call *excess-over-self predictability* (XSP): the gain in one-step predictability of a variable when a learned low-dimensional code of the entire remaining system is added to a flexible model of the variable's own history. The code comes from a small ensemble of unsupervised masked autoencoders; each per-variable readout is a ridge regression, so a complete scan of 71,721 variables runs in minutes on a single consumer laptop GPU: a complete whole-brain scan of a 71,721-neuron zebrafish recording costs under four minutes of training and roughly ten minutes of readout, against the $\approx$31,700 compute-hours a pairwise scan of comparable width would require. This asymmetry, four orders of magnitude, is what makes the question askable at all on ordinary hardware. Every analysis embeds a *ghost channel*, a circularly shifted copy of a real variable whose score must be zero, giving each dataset a built-in falsification test that needs no ground truth. On synthetic systems with known membership the statistic attains top-10 precision of 30/30 across three system seeds, with non-member excess never exceeding $+0.0005$ across the $\sim$900 autonomous channels of each system. Across four open real-world domains the statistic behaves as theory predicts, and fails informatively where it does not. In *C. elegans* it ranks the canonical motor-command circuit as the brain's driven core without access to anatomical or functional labels (top-10 all command/motor in two of three analysis-development worms, the command core topping both held-out rankings; all predictions stated in advance pass in five of five), and it tracks a genetic silencing of the command hub AVA neuron-by-neuron. In a 71,721-neuron zebrafish recording the driven core is spatially coherent (1.78$\times$ tighter than chance) and posterior-midline. In clinical electroencephalography (EEG; the statistic maps seizure spread, not onset), seizures measurably concentrate network drive, while per-event spread maps fail to replicate, a negative result with practical implications for single-event connectivity mapping. In 77 years of daily sea-level pressure the driven core is the deep

tropics (99% of top cells within 30° of the equator), and an embedded solar-forcing channel scores zero within the ghost tolerance (-0.003), confirming the statistic's intended insensitivity to exogenous driving sources. We state semantics and limits precisely (drivenness only, top-k precision, validated down to $n \approx 2{,}000$ samples), and release code, pre-registrations, and a ledger including all negative results.

1 Introduction

Causal structure discovery from time series is well served by established tools at small scale. Convergent cross mapping exploits Takens' embedding theorem to detect coupling between pairs of variables on shared attractors [1, 2, 36]; constraint-based frameworks such as PCMCI (the Peter–Clark step combined with a momentary-conditional-independence test) condition away spurious associations across moderate variable counts [3, 25]; and Granger causality and transfer entropy define the underlying conditional-predictability quantity [4, 5]. The question increasingly posed in practice, however, is which of ten thousand observed variables the system drives. That question lies outside the demonstrated operating range of every existing tool. Neural implementations of Granger causality report tens to $\sim$100 variables [6], and amortised causal discovery more broadly remains demonstrated at comparable scale [38]; our own wall-clock measurement of pairwise CCM under the full protocol ($\sim$1.1 s per pair at n=2,000) projects a single 10^4-variable scan at roughly 31,700 compute-hours ($\approx$3.6 years); and the natural accelerations fail in our tests (cross-mapping against principal-component summaries scores its own control at the maximum; kNN conditioning [29] collapses at 18 conditioning dimensions in a two-seed synthetic test).

No existing tool closes this. Our own measurements above leave pairwise CCM wall-clock infeasible at this width, and its natural acceleration via kNN conditioning [29] sample-starved by 18 dimensions; PCMCI [3] is untested at this width and, on the synthetic chains examined later in this paper (Section 6), acquires mediated false positives near synchrony rather than degrading gracefully; and the closest learned relatives, neural Granger causality [6] and amortised causal discovery more broadly [38], stop at tens to $\sim$100 variables and ship no embedded, ground-truth-free validity check of their own. **No existing method estimates variable-level drivenness at $V \gtrsim 10^3$ with a built-in, ground-truth-free validity check; this is the gap this paper closes.**

This paper closes that gap. What is new is not the causal quantity: Granger causality and transfer entropy [4, 5, 28] already define what excess-over-self predictability measures, masked autoencoders [8, 39] are an existing representation-learning tool rather than one introduced here, and the ghost channel is a direct descendant of surrogate-data testing [21]. The novelty is in what makes that existing quantity usable at $V \gtrsim 10^3$, in three parts. **(i)** We isolate, with separating experimental signatures rather than an aggregate accuracy drop, three distinct mechanisms by which learned causal readouts fail on redundant dynamical systems: maturity collapse of difference-based importance, typicality substitution in masked reconstruction, and ensembling's inability to remove shared bias (Section 2). **(ii)** These mechanisms jointly dictate, rather than merely motivate, a single corrected statistic: an amortised Granger/transfer-entropy-type quantity whose self-baseline cancels typicality by construction and whose embedded ghost channel supplies the built-in, ground-truth-free validity check that closes the gap above. The statistic is amortised, via unsupervised compression, to the tens of thousands of variables demonstrated later in this paper, where the classical and neural incumbents above are wall-clock infeasible, sample-starved, mediation-prone, or undemonstrated (Section 3); we validate it where membership is known by construction (Section 4). **(iii)** Beyond synthetic benchmarks, we deploy the statistic across four open real-world domains at a cost small enough to place the method within reach of any laboratory with a workstation, and subject it to a stronger evidence class than correlational validation alone: an interventional test in which the statistic, computed purely observationally, tracks a targeted genetic silencing of the *C. elegans* command hub AVA neuron-by-neuron in directions stated before the data was opened, alongside a measured method-selection map (Sections 5–6).

Terminology across fields

This work spans dynamical systems, machine learning, neuroscience, clinical electrophysiology and climate science; we define the recurring terms once, here. *Driven / drivenness*: a variable is driven if the state of the rest of the system carries information about its next value beyond the variable's own history; its degree is the excess statistic of Eq. 1. *Autonomous*: evolving under its own dynamics only; the opposite of driven. *Source (exogenous root)*: a variable that influences others but is driven by nothing observed; invisible to the statistic by design. *Delay embedding*: the vector of a variable's recent lagged values, which for deterministic systems reconstructs the underlying state [2]. *Ghost channel*: a circularly time-shifted copy of a real variable, embedded in the analysis as a negative control. *Masked autoencoder*: a neural network trained to restore randomly hidden inputs from the remainder, yielding a compressed *code* of the whole system. *Ensemble / consensus*: several identically trained networks differing only in random initialisation, and the average of their readouts. *Self-baseline*: the predictability of a variable from its own history alone, subtracted in Eq. 1. *Top-k precision*: the fraction of the k highest-ranked variables that are true positives. *Average precision (AP)*: the area under the precision-recall curve; its chance level equals the positive-class fraction (the *baseline*). *AUROC*: area under the receiver-operating-characteristic curve; 0.5 is chance. *Ictal / interictal*: during / between epileptic seizures. *Connectome*: the anatomical wiring diagram of a nervous system. *Reanalysis*: a historical gridded reconstruction of the atmosphere from observations. *Teleconnection*: a statistical linkage between climate variability in distant regions. *Pre-registration*: fixing predictions and analysis choices in a written artifact before inspecting the data.

Registration discipline varied by study and we state it exactly: the EEG and climate deployments were registered in a protocol document before their data was accessed; the wild-type worm predictions were fixed in the analysis script's header before execution, with two animals reserved untouched for held-out confirmation; the intervention directions were stated in the working ledger before the silenced-worm data was opened; the zebrafish analysis was exploratory apart from its in-code ghost gate. Negatives are reported at the same volume as positives, and every number below is reproducible from the released code and public data.

2 Three failure mechanisms

All experiments in this section use coupled logistic-map networks (with heterogeneous and stochastic variants) where ground truth is known by construction; full protocols are in the released ledger. Every analysis embeds a *ghost channel*, our standing negative control, in the tradition of surrogate-data hypothesis testing [21, 22]. The ghost is built by taking one real variable's series and rotating it in time by a large random offset: the rotated copy retains every single-channel property of real data—the same distribution of values, the same autocorrelation,

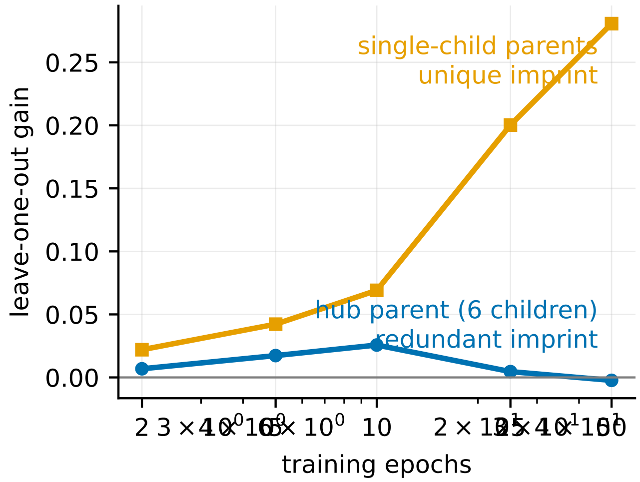

Figure 1: **Maturity collapse is redundancy-ordered, not uniform.** Leave-one-out gain versus training epochs on a hub-structured synthetic network (mean of two system seeds). Gains for the hub parent, whose imprint is redundant across six children, rise, peak at partial maturity, and collapse to zero as the model learns alternative routes; gains for single-child parents, whose imprint is unique, grow monotonically. Uniform degradation—the overfitting signature—is absent.

the same spectrum—but its moments no longer line up with anyone else's, so it is, by construction, causally connected to nothing. A valid analysis must therefore score it at zero; any nonzero ghost score measures the analysis's own artifact level. This is the logic of surrogate-data testing, embedded as a channel and scored by the same machinery as every real variable.

Mechanism 1 (Maturity collapse of difference-based importance)**.** *Difference-based scores (leave-one-covariate-out and relatives) measure a trained model's loss when one input is removed. When a cause's imprint is redundant across several observed effects (generic for dynamical systems, since any single effect's delay embedding reconstructs its driver [2]), the difference collapses to zero* precisely when training succeeds*: a mature model re-routes around any single removed input. The separating signature, which distinguishes this collapse from overfitting, is the following. On a hub-structured network, checkpointing each model across its own training trajectory shows redundant-parent scores rising, peaking at partial maturity, and collapsing toward zero by epoch 50 in the seed mean (complete in one seed, partial in the other). Unique-parent scores grow monotonically throughout in both seeds (*$0.02 \to 0.28$*). The crossing of these two curves is the synthetic signature. On real neural data (*C. elegans*, dense connectome redundancy [10, 11]) the collapse endpoint replicates: mature leave-one-out gains are exactly zero—AUROC* 0.500 *to three decimals in all three worms tested, and no intermediate checkpoint yields informative scores. This is the function-class dependence of importance gaps [7] expressed along the training trajectory: such scores work* because of *restricted capacity, not despite it. Figure 1 shows the crossing.*

Mechanism 2 (Typicality substitution in masked reconstruction)**.** *Scoring a channel by how well a trained masked autoencoder [8] reconstructs it from the rest is a natural foundation-model-style probe of coupling [8, 39]. At scale, on homogeneous pools, it inverts: on a 1,000-channel*

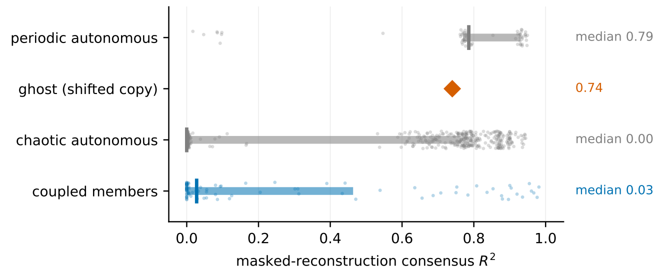

Figure 2: **Masked reconstruction measures typicality, not coupling.** Consensus masked-reconstruction R^2 per channel class at $V{=}1001$ (8-encoder ensemble). Periodic autonomous channels and the ghost—which carries no causal information by construction—score highest; genuinely coupled members score lowest. The causal ordering is inverted.

benchmark (a coupled minority of ∼100 among ∼900 autonomous channels), reconstruction ranks periodic autonomous channels highest (0.76), the ghost at 0.74, chaotic autonomous channels at 0.29, and true members last (0.25). The decoder's loss-optimal completion of a masked slot is the family-typical *trajectory, so reconstruction measures typicality, not coupling. Coupled channels, perturbed off the family manifold by their drive, are the least typical. Bottleneck width (*$8 \to 256$*) and masked-only losses fail to repair the inversion, because neither changes the optimal blind completion (fill-value ablations, run at* $V{=}30$ *where no inversion exists, showed no effect there). Figure 2 shows the inversion.*

Mechanism 3 (Ensembles average variance, not bias)**.** *A natural remedy for a learned readout's failures is ensembling [27]: training many models from independent initialisations and using their consensus. We tested this directly: eight masked autoencoders on one fixed system, only the seed varying. The consensus average-precision curve rises with ensemble size (*$0.111 \to 0.148$*) with the expected* $1/\sqrt{M}$ *deceleration, exactly as the law of large numbers predicts, and converges to a biased limit: the ghost's reconstruction score is* 0.740 ± 0.004 across the eight independent minima. *The typicality artifact is a property of data-plus-objective, reproduced in every basin; averaging over minima cannot remove it. Figure 3 shows both curves.*

3 The excess-over-self statistic

The three mechanisms jointly motivate a single correction. Let $x_q(t)$ be variable q after per-variable standardisation and first-differencing (stationarity is a prerequisite; on drifting data the circular-shift ghost inflates and the scan must be discarded rather than thresholded, and the ghost must be checked, since it does not announce failure). Form delay embeddings ($E = 3$ throughout) and the joint state $Z(t)$.

We call the complete procedure (the encoder ensemble, the excess-over-self readout, and the embedded ghost control) **GhostScan**, and the statistic it computes excess-over-self predictability (XSP).

Code. Train M masked autoencoders (independent initialisations) on Z: random channel subsets are corrupted each batch, the loss is computed on masked positions only [8], and a hard linear bottleneck of width $b \ll \dim Z$ yields codes $c_m(t)$.

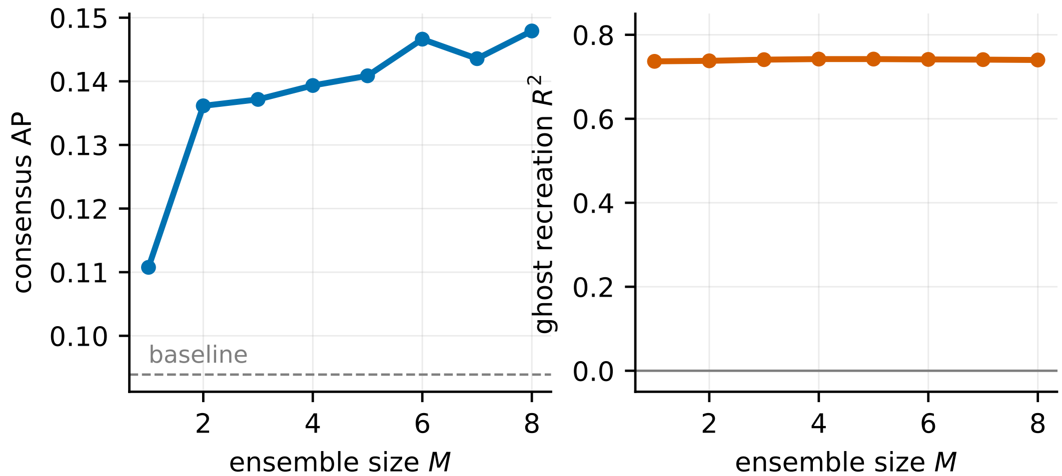

Figure 3: **Ensembling averages variance, not bias.** Left: consensus average precision rises with ensemble size exactly as the law of large numbers predicts, converging toward ∼0.15—signal plus shared bias, far below the precision required for detection (dashed line: baseline precision). Right: the ghost channel's reconstruction score across the same prefix ensembles is 0.740 ± 0.004: the artifact is reproduced in every independent minimum.

Readout. For each variable q and model m,

$$\text{excess}_m(q) \;=\; R^2\big[\,x_q(t{+}1) \;\big|\; \phi\big(\text{own lags of } q\big) \oplus c_m(t)\big] \;-\; R^2\big[\,x_q(t{+}1) \;\big|\; \phi\big(\text{own lags of } q\big)\big]\,, \quad (1)$$

with both terms ridge regressions fit on a training segment and evaluated on a held-out contiguous tail (R^2 clamped at zero), and ϕ a polynomial feature map (degree 2 on the quadratic-map synthetics, where it is exact; degree 3 on all real data; both degrees shown on the heterogeneous pool). The reported score is the consensus mean over models. The subtraction cancels typicality by construction (a channel predictable from its own structure gains nothing from the code); the statistic is not a difference across redundant alternatives, so Mechanism 1 does not apply (the architecture-identical absolute readout was empirically stable across 5–100 encoder-training epochs at $V{=}30$; the excess readout reuses those saved encoders); and its lineage is explicit: a Granger/transfer-entropy-type quantity [4, 5, 28] — the two are equivalent for Gaussian variables [28] amortised to large V by unsupervised compression [37]—CCM's convergence-with-library-size criterion makes the analogous self-control implicitly [1]; Eq. 1 makes it explicit and scalable.

Semantics. Positive excess means the system's state carries information about X's next step beyond X's own past: X is *driven*. We call this property *drivenness*, a variable's rank under the statistic its drivenness percentile (0–1 scale), and the top-k variables by consensus excess a system's *driven core*. A pure source is perfectly predicted by its own history and is therefore invisible. This is a designed, testable asymmetry: known-exogenous variables must score zero (verified below on solar forcing; a genetic silencing approximates it, since the silenced hub falls sharply but not to zero, retaining residual input).

Table 1: **Measured cost of a complete GhostScan analysis**, on one consumer laptop (RTX 3070, 8 GB). Training times are from archived model timestamps; readout is a ridge fit per variable. The final column is the cost of the pairwise alternative at the same width, from our own wall-clock measurement of CCM (∼1.1 s per pair).

deployment	V	encoder training	full readout	pairwise equivalent
worm (per animal)	∼130	<1 min	seconds	∼5 h
EEG (per window)	24	seconds	seconds	minutes
climate	10,512	∼5 min	∼16 min	≈17,000 h
zebrafish	71,721	∼4 min	∼10 min	≈800,000 h

The self-model's function class is critical. With a linear self-baseline the correction fails (a linear fit of a quadratic map leaves residual self-structure that the code then "explains", scoring periodic channels +0.26); with polynomial features matched to the dynamics (degree 2 suffices on the quadratic maps), self-baselines match theory to three decimals and the confound is cancelled rather than suppressed. On real data, where no exact function class exists, the ghost and known-root channels are the practical check.

Cost, and why it is the practical argument. Every result in this paper was produced on one consumer laptop with a single RTX 3070 GPU (8 GB) and 28 GB of system memory. No cluster, no multi-GPU node, and no cloud budget was used at any point. Table 1 gives the measured wall-clock costs.

Three properties produce this. First, the encoder is trained *once* per system and is unsupervised, so its cost is independent of how many variables one subsequently queries. Second, each per-variable readout is a ridge regression on a handful of features, at ∼8 ms per variable, so scoring scales linearly in V rather than quadratically. Third, saved encoder weights make any new readout (a different self-baseline, a new statistic, a re-analysis under a reviewer's suggestion) an evaluation rather than a retrain, at seconds per variable-sweep.

The consequence is worth stating plainly, because it determines who can use this. A pairwise scan of the zebrafish recording would require roughly ninety GPU-years; the same question, asked through GhostScan, took fourteen minutes on a laptop that also ran the authors' text editor. The method is not merely faster than the alternative at scale, it is the difference between an analysis that a small laboratory can run on existing hardware and one that no laboratory would attempt.

4 Validation on systems with known ground truth

We validate on systems of $V \in \{30, 1000\}$ logistic-map channels in which a minority sub-web is coupled (sparse random DAG) and the remaining channels are autonomous; membership is known by construction. At V=1001 (93–95 members by seed; baseline precision ≈0.094):

system	precision@10	precision@20	false alarms (907 non-members)
seed 0	10/10	16/20	0
seed 1	10/10	20/20	0
seed 2	10/10	20/20	0
heterogeneous pool	10/10	18/20	0

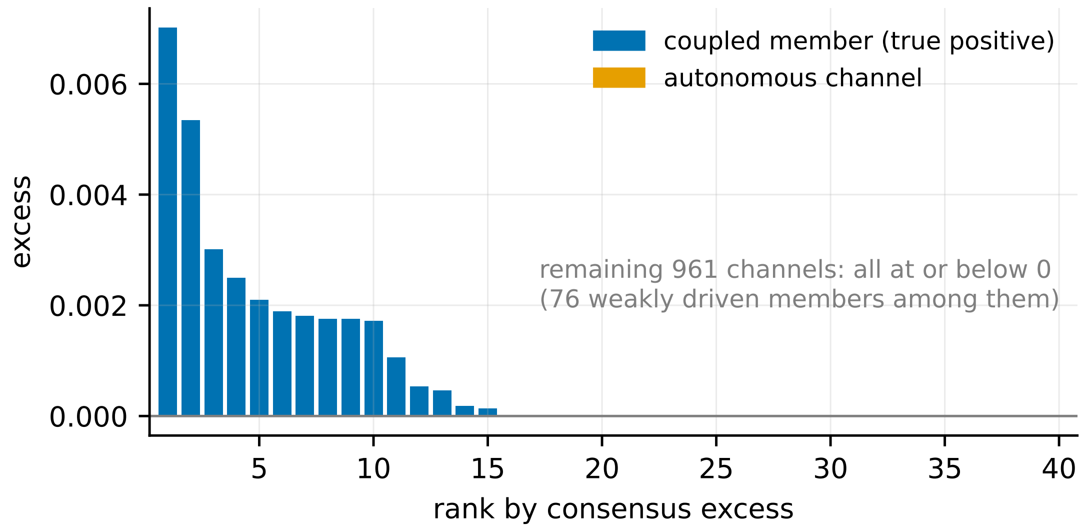

Figure 4: **The excess statistic at V=1001 (system seed 0).** Channels ranked by consensus excess. The top of the ranking is pure coupled members (blue); all 907 autonomous channels (orange) and the ghost (star) pin at or below zero. The statistic is a top-k detector: weakly driven members fall below the zero-pinned non-members, which is why full-population AUROC (0.28) is uninformative while top-10 precision is perfect.

All autonomous channels—including 52–71 periodic ones per system, a deliberately stochastic AR(1) family in the heterogeneous pool, and the ghost, lie at or below zero excess (Figure 4).

Two boundaries of the statistic were measured directly. First, it is a **top-k detector** (members whose drive share falls below the readout's noise floor receive slightly negative excess and rank below the zero-pinned non-members, so full-population AUROC is uninformative even when top-k precision is perfect—0.28 here by mechanism, not accident; average precision and top-k against stated baselines are the correct metrics). Second, the **encoder is necessary** (the same readout conditioning on all raw dimensions with tuned regularisation fails, AP 0.10 vs. 0.24). Third, **ensembling is helpful but not critical** (archived single-model AP 0.25 vs. consensus 0.24; its chief value is the cross-model variance diagnostic). "False alarm" throughout means non-member excess above the ghost floor; the largest non-member excess observed in any synthetic system is +0.0005, an order of magnitude below the weakest top-20 member.

5 Real-world validation and deployments

We deploy the statistic on four open real-world domains. All datasets are public and unrestricted. Registration status is stated per study below; the released protocol document and analysis-script headers are the citable artifacts. Table 2 summarises the validity gates across all deployments.

5.1 *C. elegans*: label-free recovery of the motor-command circuit, and an interventional test

Wild type. Whole-brain calcium imaging of five immobilised worms (109–135 neurons, ∼3,100 samples) [9]. Predictions fixed in the analysis script before execution: ghost ≈ 0; sensory neurons

Table 2: **Validity gates across deployments.** The ghost is a circularly shifted real channel (must score ≈ 0); exogenous-root channels are known non-driven variables (must score ≈ 0).

deployment	V	n	ghost	exogenous root
synthetic ($\times$3 seeds + het.)	1,001	2,000	≤ 0	AR(1) family ≈ 0
worm, wild type ($\times$5)	109–135	$\sim$3,100	-0.03 to 0.00	—
worm, AVA silenced ($\times$5)	120–138	$\sim$3,000	-0.02 to -0.01	silenced hub falls to percentile 0.38
zebrafish	71,721	2,241	-0.031	—
EEG (chb01, 14 windows)	23	$\geq$2,000	$\lvert\cdot\rvert \leq 0.034$	—
climate	10,512	$\sim$28,000	-0.005	sunspots -0.003

(hand-curated canonical classes; n=8/6/1 per development worm, so per-worm evidence is weak though the direction is uniform) less driven than command/motor neurons in the constant environment; positive correlation with anatomical in-weight [10, 11]. All three passed on the three animals used during analysis development (hereafter *development worms*) and again on two animals reserved since the study's declaration and opened only after predictions were fixed, with the strongest anatomy correlations of the five (+0.30, +0.34). The top-10 most-driven neurons are 10/10 command/motor class in two of the three development worms, with the command core topping the two held-out rankings as well: AVA, AVE, RIM, AIB, RIB, RIV, RME, SMDV and ventral-cord motor neurons. This is the canonical motor-command ensemble of the immobilised worm [9], obtained without access to anatomical or functional annotations.

Intervention. The same pipeline on five worms with the reverse-command hub AVA silenced via a histamine-gated channel [9]. Directions stated in the working ledger before the silenced-worm data was opened: the silenced hub shows the largest decrease within the command core (drivenness percentile 0.92$\to$0.38); its module and motor pool collapse (AVE -0.28, RIB -0.29, VB01 -0.64, RMED -0.49, VA01 -0.45); ghost scores remain at zero in all five animals. Two further observations emerge from the pooled per-cell contrast, and both deserve unpacking rather than a bare number.

RIM and AIB retain drive almost unchanged (-0.08, -0.08, class-pooled across the five-animal cohorts, though not uniformly in every single cell, since one silenced animal's RIMR sits low while its RIML tops the ranking) while every other module member collapses. This is mechanistically notable rather than incidental. Sordillo and Bargmann established that RIM is not a passive relay of upstream command input but carries its own bistable depolarised/hyperpolarised dynamic that shapes reversal behaviour on its own terms [13], and Piggott et al. document a strong AIB$\to$RIM synaptic route capable of operating independently of AVA's own command synapses [30]. A network-drivenness score for this pair that survives near-intact when the circuit's dominant hub is silenced is the pattern that anatomy predicts *if* RIM's intrinsic dynamics and the AIB$\to$RIM route are supplying the residual drive. The identical pattern would also arise from a readout that is simply insensitive to whatever we manipulate, and five animals under one perturbation cannot distinguish the two. Confirming the first reading over the second would need independent replication, a manipulation that targets AIB or RIM directly rather than routing through AVA, and single-cell (not class-pooled) consistency, which the RIML/RIMR split above already shows is not yet in hand.

Drive reorganises onto the forward hub AVB (+0.34; AVB is recorded in three of the five silenced animals and the rise is carried by two of them, so this observation is the weaker of the pair; ALA rises by a comparable margin, +0.30, Table 3, but, unlike RIM, AIB and

Table 3: **Drivenness percentile, wild type (WT) versus AVA-silenced** (AVA:HisCl, histamine-gated chloride channel; cells identified in ≥ 3 worms of both five-animal cohorts; L/R-collapsed; percentiles on a 0–1 scale).

cell	WT	AVA:HisCl	Δ
AVA (silenced hub)	0.92	0.38	−0.54
VB01 (motor)	0.91	0.28	−0.64
RMED (motor)	0.90	0.41	−0.49
VA01 (motor)	0.82	0.37	−0.45
AVE	0.97	0.69	−0.28
RIB	0.66	0.37	−0.29
RIM	0.93	0.84	−0.08
AIB	0.75	0.65	−0.09
AVB (forward hub)	0.49	0.82	+0.34
ALA	0.16	0.46	+0.30

AVB, has no established place in the AVA/AVB/AIB/RIM reversal-command architecture in the sources we cite, so we report it in the table and leave it uninterpreted rather than assign it a mechanism we cannot support). AVB is the forward-command counterpart to AVA within the same architecture, driving forward locomotion as AVA drives reversal, and the same whole-brain recordings that motivated the wild-type analysis established that this worm's global dynamics organise into a small number of alternating command states rather than one fixed hierarchy [9]. A rise in AVB's own drivenness once the reversal hub is silenced is the pattern that organisation predicts if the rest of the network reorganises around whichever command pathway remains rather than simply losing an input channel. With the signal carried by only two of three recorded animals, however, a small-sample artifact is at least as economical an explanation, which is why we call this the weaker of the two observations.

We report both as candidate observations pending adjudication against the reversal-circuit literature [12, 13]; independent of whether these observations are novel, an observational statistic tracked a targeted genetic intervention neuron-by-neuron in the stated directions. Table 3 and Figure 5 give the per-cell contrast.

5.2 Zebrafish: a whole-vertebrate-brain map

ZAPBench provides 71,721 segmented neurons $\times$ 7,879 timesteps of light-sheet imaging in a larval zebrafish [14]. On a 2,241-step window (open-loop, rotation, dark), the full-brain excess map is computed in minutes on a laptop GPU. The ghost scores -0.031; $\sim$2,800 neurons clear the $+0.01$ excess threshold; and the driven core is spatially coherent: the top-1,000 neurons are 1.78$\times$ more tightly packed than random draws (median nearest-neighbour distance 16.8 vs. 29.8 ± 0.6), concentrated posteriorly and against the midline, consistent with the hindbrain/midline territory where internally generated motor dynamics are reported to reside [23]—in a window dominated by the dark condition. Region names await atlas registration and are not claimed (Figure 6).

5.3 Clinical EEG: a confirmed effect and an instructive null

CHB-MIT scalp EEG, subject chb01: 23 channels, 256 Hz; six of the seven annotated seizures are analysed (the seventh record was not in our retrieved subset) [15, 16], in a field where seizures

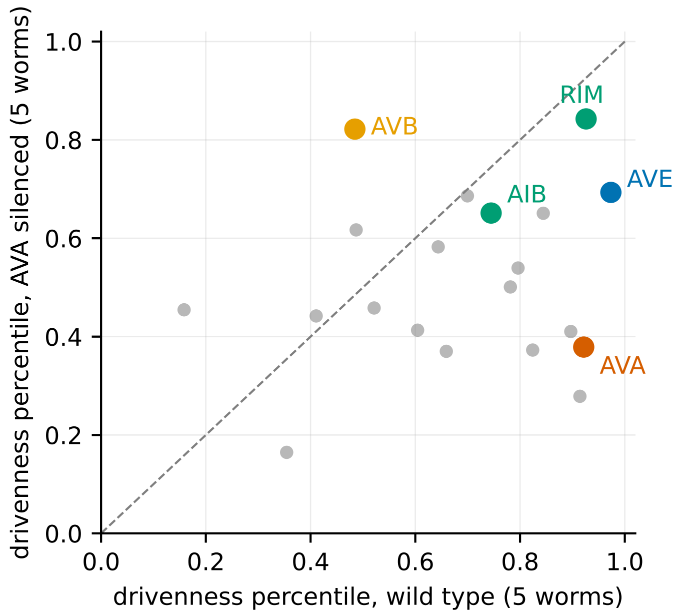

Figure 5: **An observational statistic tracks a genetic intervention.** Per-cell drivenness percentile in wild type (horizontal) versus AVA-silenced animals (vertical); each point is a cell identified in at least three worms of both cohorts. The silenced hub AVA and its module fall below the diagonal; RIM and AIB barely move; the forward hub AVB rises above it.

are increasingly understood as network-organisation disorders [33–35]. One caveat governs all outputs: seizure-onset zones are *sources*, and sources are invisible to this statistic by design. The statistic maps the spread network, not the origin. Results: max |ghost| 0.034 across the 14 windows (no numeric tolerance was pre-specified; ghost magnitudes across the other deployments range from ∼0.000 on the synthetics to 0.041 on one wild-type worm); drivenness *concentration* (top-4 channel share) is higher during seizures than between them (0.68 vs. 0.47; five of six within-record pairs, with the exception, record 03, in the opposite direction). Seizures therefore recruit the network into a driven regime (Figure 7).

The third pre-registered prediction failed, and this failure is itself informative: per-event driven patterns do not replicate across the same patient's seizures (mean pairwise Spearman 0.08 over 15 pairs). Whatever the cause—genuine per-event variability or single-window noise—the practical warning stands: single-event connectivity maps at these window lengths should not be treated as patient signatures without per-patient replication; we tested one subject.

5.4 Climate: a tropical driven core and an invisible solar forcing

Seventy-seven years of daily sea-level pressure (NCEP/NCAR Reanalysis-1; 10,512 cells, ∼28,000 days) [17], deseasonalised and differenced, with the SILSO daily sunspot number [19] embedded as an additional channel. Pre-registered gates: ghost −0.005; **sunspot channel** −0.003—a real exogenous forcing, embedded among ten thousand atmospheric variables, scores nothing: the cleanest field test of the statistic's designed insensitivity to exogenous sources; and the driven core is spatially clustered (top-500 median nearest-neighbour distance 0.0435 vs. 0.0676 ± 0.0048 under a spatial null). The exploratory map, fixed to carry no prior claim, reproduces well-

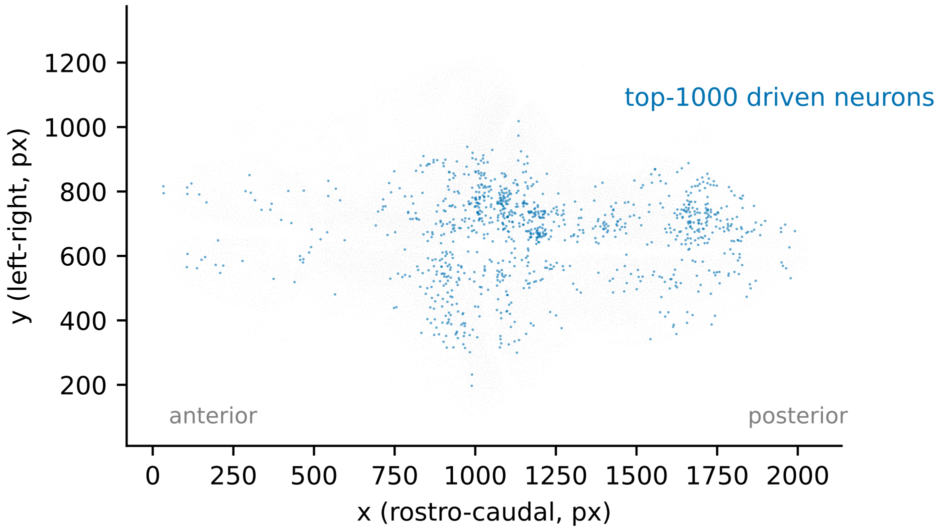

Figure 6: **The driven core of the larval zebrafish brain is spatially localised.** All 71,721 segmented neurons in horizontal projection (gray); the 1,000 highest-excess neurons (blue) concentrate posteriorly and near the midline (median nearest-neighbour distance 16.8 versus 29.8 ± 0.6 for random draws of 1,000).

established large-scale structure: 99% of the 500 most-driven cells lie within 30° of the equator (mean |lat| 6.1° vs. 45.6° for the grid). This is the Walker/ENSO belt [31, 32], where tropical pressure is subordinate to planetary-scale circulation [32], while polar and storm-track cells, dominated by locally generated baroclinic dynamics [26], rank most autonomous. Every cell shows positive excess; a single connected fluid contains no autonomous members (Figure 8).

6 An empirical method-selection map

Because the goal is a usable tool, we summarise the regimes in which each method performed reliably in our measurements, including where ours should not be used.

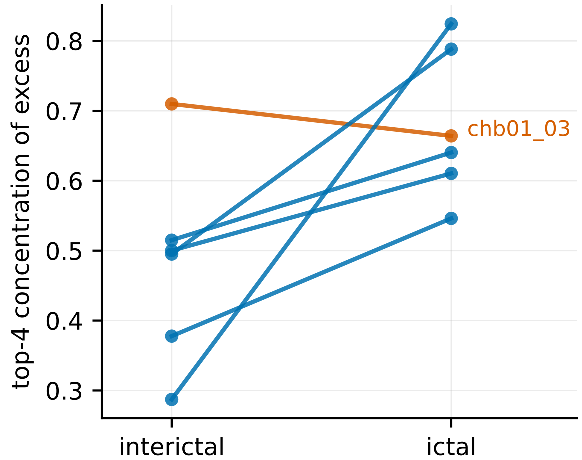

Figure 7: **Seizures concentrate network drive.** Top-4 channel share of total excess, interictal versus ictal, per record with an annotated seizure. Five of six records move in the predicted direction; the exception (record 03, red) is included.

regime	recommendation	basis (measured)
pairs / small systems, weak coupling	CCM	detects coupling 0.01 in seconds; learned methods need 0.04–0.08
short real series ($n \lesssim 50$)	conditional kNN first (best AP, 1.75× baseline); CCM and PCMCI beside it	measured on gene-circuit data [18]; no trained estimator viable
noise-driven, near-linear systems	PCMCI or CCM	statistically indistinguishable on wind-tunnel physics [20] (AP 0.265 vs. 0.256); learned reconstruction methods fail there and emit false positives on exogenous actuators
strong coupling near synchrony	conditional learned models (*candidate only*: synthetic chains, 3 seeds, unverified; did not transfer to the one dense real circuit tested)	CCM, conditional kNN and PCMCI all acquire mediated false positives from coupling 0.3–0.5 on synthetic chains
membership at $V \gtrsim 10^3$	this statistic	pairwise CCM is infeasible in our wall-clock measurements and kNN conditioning is sample-starved; PCMCI at this width is untested; top-k precision with ghost control
edge-level maps on dense real circuits	no method we tested	difference-based scores collapse under redundancy (Mechanism 1)

Absolute expectations matter as much as rankings: on real data the best methods deliver roughly 1.5–2.4× baseline average precision—far from clean recovery.

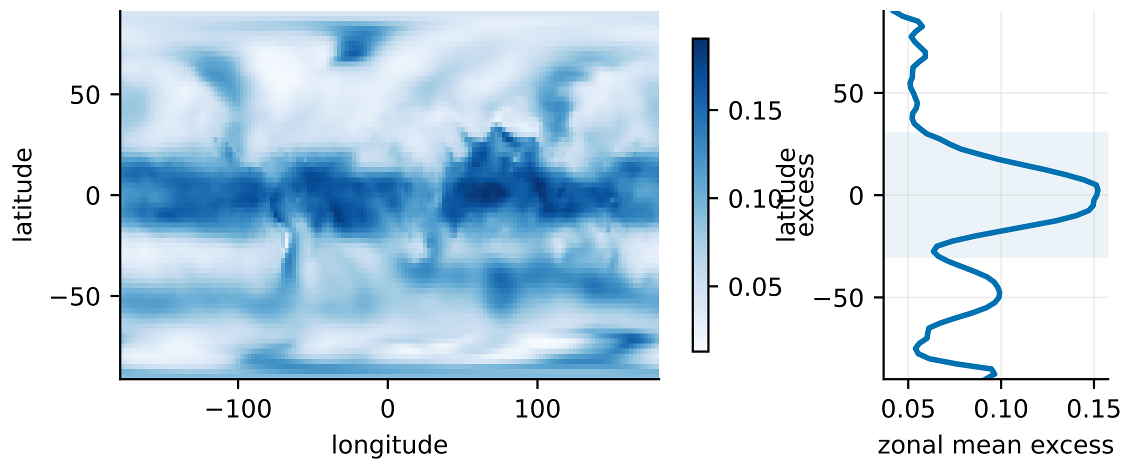

Figure 8: **The atmosphere's driven core is the deep tropics.** Left: excess per grid cell for 77 years of daily sea-level pressure (single-hue scale; darker = more driven). Right: zonal mean excess by latitude; the shaded band marks ±30°, which contains 99% of the 500 most-driven cells. Polar and mid-latitude storm-track cells—dominated by locally generated dynamics—rank most autonomous.

7 Limitations

(1) Drivenness only; sources are invisible, and a source-detection complement is future work. (2) Top-k semantics; weakly driven members are indistinguishable from non-members. (3) Validity is established down to $n \approx 2{,}000$ samples; no learned variant we tested succeeded on a 48-point circadian dataset, and the intermediate regime is unmeasured. (4) Stationarity is an essential assumption. (5) Single recordings per condition in the biological analyses; the intervention contrast is five animals per cohort from one preparation; the EEG analysis covers one subject; absolute excess magnitudes on the worm are small (top-band 0.02–0.23 across cohorts); synthetic validation is confined to map-type generator families. (6) The self-model's function class governs transfer; the ghost and known-root channels are the practical guard. (7) The statistic estimates a known causal quantity; the contribution is scale, controls, failure analysis and validation, not the concept.

8 Conclusion

Three failure mechanisms explain why learned causal readouts break on redundant dynamical systems. The correction they imply, excess over self-predictability computed against a compressed code of everything else, turns chance-level detection into high-precision membership scans. It does so at widths where pairwise scanning is infeasible in our wall-clock measurements and local conditioning is sample-starved, and it behaves as theory predicts across worm, fish, clinical and geophysical data, including under a genetic intervention. The operational protocol that found every failure here (embedded ghosts, known-root channels, trajectory checkpointing, pre-registration, precision-recall against stated baselines, published negatives) is, we believe, as transferable as the statistic itself.

Data and code. All datasets are public and unrestricted (OSF, ZAPBench, PhysioNet, NOAA PSL, SILSO, NCBI GEO, causal-chamber releases, OpenWorm). All code, the complete experiment ledger including negative results and voided attempts, the pre-registration documents, and the scripts that generate every figure in this paper from the primary data are available at `https://github.com/AkandaAshraf/DeepFeatSelection`. Code, pre-registration documents, and the complete experiment ledger— including negative results and voided attempts—are released alongside this paper.

References

[1] G. Sugihara et al. Detecting causality in complex ecosystems. *Science* 338:496–500, 2012.

[2] F. Takens. Detecting strange attractors in turbulence. *Lecture Notes in Mathematics* 898:366–381, 1981.

[3] J. Runge et al. Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances* 5:eaau4996, 2019.

[4] C. W. J. Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 37:424–438, 1969.

[5] T. Schreiber. Measuring information transfer. *Physical Review Letters* 85:461–464, 2000.

[6] A. Tank, I. Covert, N. Foti, A. Shojaie, E. Fox. Neural Granger causality. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44:4267–4279, 2022.

[7] Y. Xu, S. Zhao, J. Song, R. Stewart, S. Ermon. A theory of usable information under computational constraints. *ICLR*, 2020.

[8] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick. Masked autoencoders are scalable vision learners. *CVPR*, 2022.

[9] S. Kato, H. S. Kaplan, T. Schrödel, S. Skora, T. H. Lindsay, E. Yemini, S. Lockery, M. Zimmer. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell* 163:656–669, 2015.

[10] J. G. White, E. Southgate, J. N. Thomson, S. Brenner. The structure of the nervous system of the nematode *C. elegans*. *Philosophical Transactions of the Royal Society B* 314:1–340, 1986.

[11] S. J. Cook et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature* 571:63–71, 2019.

[12] J. M. Gray, J. J. Hill, C. I. Bargmann. A circuit for navigation in *Caenorhabditis elegans*. *PNAS* 102:3184–3191, 2005.

[13] A. Sordillo, C. I. Bargmann. Behavioral control by depolarized and hyperpolarized states of an integrating neuron. *eLife* 10:e67723, 2021.

[14] J.-M. Lueckmann et al. ZAPBench: a benchmark for whole-brain activity prediction in zebrafish. *ICLR*, 2025.

[15] A. Shoeb. Application of machine learning to epileptic seizure onset detection and treatment. PhD thesis, MIT, 2009.

[16] A. L. Goldberger et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* 101:e215–e220, 2000.

[17] E. Kalnay et al. The NCEP/NCAR 40-year reanalysis project. *Bulletin of the American Meteorological Society* 77:437–471, 1996.

[18] M. E. Hughes et al. Harmonics of circadian gene transcription in mammals. *PLoS Genetics* 5:e1000442, 2009.

[19] SILSO World Data Center. The international sunspot number. Royal Observatory of Belgium.

[20] J. L. Gamella, J. Peters, P. Bühlmann. Causal chambers as a real-world physical testbed for AI methodology. *Nature Machine Intelligence* 7:107–118, 2025.

[21] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, J. D. Farmer. Testing for nonlinearity in time series: the method of surrogate data. *Physica D* 58:77–94, 1992.

[22] W. Ebisuzaki. A method to estimate the statistical significance of a correlation when the data are serially correlated. *Journal of Climate* 10:2147–2153, 1997.

[23] T. W. Dunn, Y. Mu, S. Narayan, O. Randlett, E. A. Naumann, C.-T. Yang, A. F. Schier, J. Freeman, F. Engert, M. B. Ahrens. Brain-wide mapping of neural activity controlling zebrafish exploratory locomotion. *eLife* 5:e12741, 2016.

[24] M. B. Ahrens, M. B. Orger, D. N. Robson, J. M. Li, P. J. Keller. Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nature Methods* 10:413–420, 2013.

[25] J. Runge et al. Inferring causation from time series in Earth system sciences. *Nature Communications* 10:2553, 2019.

[26] E. K. M. Chang, S. Lee, K. L. Swanson. Storm track dynamics. *Journal of Climate* 15:2163–2183, 2002.

[27] B. Lakshminarayanan, A. Pritzel, C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *NeurIPS*, 2017.

[28] L. Barnett, A. B. Barrett, A. K. Seth. Granger causality and transfer entropy are equivalent for Gaussian variables. *Physical Review Letters* 103:238701, 2009.

[29] A. Kraskov, H. Stögbauer, P. Grassberger. Estimating mutual information. *Physical Review E* 69:066138, 2004.

[30] B. J. Piggott, J. Liu, Z. Feng, S. A. Wescott, X. Z. S. Xu. The neural circuits and synaptic mechanisms underlying motor initiation in *C. elegans*. *Cell* 147:922–933, 2011.

[31] J. M. Wallace, D. S. Gutzler. Teleconnections in the geopotential height field during the Northern Hemisphere winter. *Monthly Weather Review* 109:784–812, 1981.

[32] J. Bjerknes. Atmospheric teleconnections from the equatorial Pacific. *Monthly Weather Review* 97:163–172, 1969.

[33] R. Vicente, M. Wibral, M. Lindner, G. Pipa. Transfer entropy—a model-free measure of effective connectivity for the neurosciences. *Journal of Computational Neuroscience* 30:45–67, 2011.

[34] K. J. Friston. Functional and effective connectivity: a review. *Brain Connectivity* 1:13–36, 2011.

[35] M. A. Kramer, S. S. Cash. Epilepsy as a disorder of cortical network organization. *The Neuroscientist* 18:360–372, 2012.

[36] H. Ye, E. R. Deyle, L. J. Gilarranz, G. Sugihara. Distinguishing time-delayed causal interactions using convergent cross mapping. *Scientific Reports* 5:14750, 2015.

[37] D. Janzing, B. Schölkopf. Causal inference using the algorithmic Markov condition. *IEEE Transactions on Information Theory* 56:5168–5194, 2010.

[38] S. Löwe, D. Madras, R. Zemel, M. Welling. Amortized causal discovery: learning to infer causal graphs from time-series data. *CLeaR, PMLR* 177:509–525, 2022.

[39] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.